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Cross-domain technical product and operations leader with an eleven-year self-directed engineering practice, combining AI systems, analytics, business strategy, hospitality, and venture execution to turn ambiguous briefs into production-minded builds and client-ready decisions.

Professional Experience

Margadarshaka

Jan 2025 to Present

Co-Founder and Technical Lead

Builds a real-time AI voice platform for career and life counselling in Indian regional languages, with cost controls and human safeguarding built into the product architecture.

- Architected a real-time voice pipeline around a projected INR 50 per call ceiling, balancing speech recognition, model inference, response latency, and human review.
- Required human review after every AI-handled call and handed counsellors a structured case dossier for follow-up.
- Rejected a premature full build after feasibility testing exposed payer-user misalignment and counselling-liability risk.
- Set a staged validation path to test demand, delivery economics, and human-review safeguards before scaling.

AstroPatshala

Jan 2021 to Present

Co-Founder and Operations Lead

Runs operations, growth, and technology for an astrology consulting and education venture while the lead astrologer manages client delivery.

- Built AstroPatshala's operating model from the ground up, turning pricing, licensing, service packaging, market entry, and delivery into a repeatable venture system.
- Built and maintained 2 production codebases that support customer acquisition, education delivery, research publishing, and internal operations.
- Ran continuous service and content experiments in Vedic astrology and Bhrgu Nadi, turning the work into an applied research base for consulting, education, and product decisions.

Hult AI Collective

Jan 2025 to Present

Founder and President

Founded a student-run AI organisation that delivers practical learning, events, and peer collaboration at Hult.

- Founded Hult's student AI organisation and took it from operating concept to public launch, putting programming, outreach, events, treasury, and technical infrastructure in place.
- Produced the HAC x Hult Finance Hackathon in partnership with the Hult Banking Society, owning partner alignment, the event dossier, speaker outreach, workshops, and on-site delivery.
- Kept the hackathon's closing programme on schedule after confirmed external speakers withdrew by designing and delivering the replacement AI workshop at short notice.
- Expanded practical AI learning beyond club events through a guest lecture on agentic workflows and AI-assisted software development for the wider student body.
- Made HAC's work reusable and publicly legible through kartavya.tech/HAC, centralising event records, technical resources, and treasury infrastructure in one operating hub.
- Built a sponsorship and partnership pipeline through fit research, institutional and commercial partner sourcing, tailored outreach, and follow-up for events and society programmes.

Savi Hotels & Resorts, Jaipur, India

Jan 2021 to Jul 2026

Operational Lead

Led hotel operations, turnaround execution, guest experience, financial controls, and operating-system improvement across four Savi properties within SRBS Group.

- Spearheaded the first month of a leased-hotel turnaround, giving the property team a shared system map across RevPAR, occupancy, staffing, guest satisfaction, asset condition, and daily service friction.
- Reduced operational friction in month two by designing mobile-first workflows and the supporting technical architecture for front-desk control, staff coordination, and property-wide service delivery.
- Raised month-on-month occupancy above the inherited baseline by combining room revenue with complementary property uses, strengthening the property's future lease and management valuation case.
- Applied a two-month kill gate when leasing velocity and profitability remained below threshold, then returned the property with a complete offboarding record and reusable pitchbook.
- Increased repeat bookings by 30% within 6 months across 4 hotels by converting structured guest feedback into targeted operating improvements.
- Improved continuity between reception shifts through a shared handover queue for reservations, arrivals, payments, special requests, and service recovery.

- Automated cash and shift reconciliation with a 14-sheet OpenPyXL workbook, replacing manual cross-checks with an auditable front desk control process.
- Advanced qualified buyer and management leads by presenting property pitchbooks, walking clients through operating economics and turnaround evidence, and managing follow-up through the business-development cycle.

SRBS Group, Jaipur, India

Jan 2021 to Jul 2026

Associate to the Managing Director

Supported the Managing Director on hospitality acquisitions, commercial diligence, governance, legal coordination, and operational continuity across SRBS Group.

- Recommended proceeding with the approximately GBP 40,000 Shimla hotel lease after integrating market conditions, prior operator books, asset quality, downside risks, operating synergies, and valuation into the acquisition thesis.
- Executed the 8-day Shimla takeover through a room-by-room property rundown, inventory and quality verification, tenancy review, risk logging, stakeholder coordination, and documented handover to the incoming operating team.
- Built a reusable seven-part diligence framework spanning market attractiveness, operator quality, asset condition, legal exposure, commercial synergies, valuation logic, and post-transaction execution risk.
- Protected group continuity during an active operational disruption by restoring access and connectivity across visa-linked institutional accounts.
- Enabled counsel to meet a time-sensitive Supreme Court filing deadline by drafting and delivering a legal addendum for a related family matter.

Family Real Estate Business, Jaipur, India

Jan 2021 to Present

Real Estate Operations Associate

Coordinates commercial preparation, leasing, sales support, and client relationships across a family property portfolio in Jaipur.

- Advanced a 12-property portfolio of 8 apartments and 4 hostels through construction coordination, commercial preparation, sales, and leasing.
- Converted local market knowledge and client relationships into transactions spanning monthly rents of GBP 800 to GBP 3,800 and penthouse sales of GBP 70,000 to GBP 90,000.

Siya The Restaurant, SRBS Group

Jan 2025 to Jan 2026

Menu Engineering Consultant

Built menu-pricing and back-of-house decision systems for a 191-item restaurant operation within SRBS Group.

- Created the restaurant's first digital consumption dataset by OCR-digitising 2 years of handwritten kitchen registers into a SKU-level baseline for par levels, preparation forecasts, and stock variance checks.
- Improved back-of-house control by linking delivery, storage, FIFO rotation, preparation allocation, shift handover, and daily reconciliation into one operating flow.
- Repriced and repositioned a 191-item menu through a margin-times-velocity matrix that guided pricing, placement, assortment, and pruning decisions.
- Equipped restaurant staff with an auditable Excel pricing model for 191 menu items, replacing opaque judgement with transparent formulas and repeatable decision rules.

Independent Healthcare Venture (Client Anonymised)

Jun 2024 to Aug 2024

Technology and Strategy Consultant Intern

Completed a two-month healthcare venture internship spanning stakeholder discovery, market strategy, product definition, and MVP delivery.

- Translated stakeholder interviews into two decision-ready deliverables: a post-discovery market report and a phased entry plan grounded in customer segments and product thesis.
- Moved the venture from discovery to launch readiness in two months by shipping three assets: an MVP website, information architecture, and implementation roadmap.

Leadership and Teaching

Hult Founders Lab

Jan 2024 to Present

Technical Startup Mentor

Built incubator MVPs, coached two successful market launches, and advised London Tech Advisors on founding engineer decisions and MVP architecture.

Peer Instruction

Jan 2021 to Dec 2021

Student Instructor

Taught junior students three core software topics through an inventory system and single page application, turning theory into applied builds.

Selected Projects and Case Work

Software Engineering and Systems Projects

Eleven-year self-directed engineering practice begun at age 8 and developed into production full-stack, API, Linux, data, and AI systems work.

- Built and maintained more than 20 Python and JavaScript repositories spanning 15.8 GB and 367,000 files across applications, automation, analytics, and infrastructure.
- Shipped production-oriented CRM, ecommerce, revenue-management, and library systems with Flask, Django, and FastAPI, taking each from data model and API design through deployment.
- Developed low-level Linux systems fluency across seven distributions through network routing, firewall, SSH deployment, and OSI-stack diagnostics.
- Restored and deployed small-scale infrastructure for individual users, school events, and institutional environments, translating low-level diagnostics into stable working setups.
- Rebuilt algorithmic depth in Dijkstra and heuristic search through implementation exercises focused on choosing the right approach and defending its complexity trade-offs.

BWC Quantitative Portfolio Audit Engine

Research prototype: reproducible Python decision-support stack built to audit the BWC simulation portfolio without connecting analysis code to live execution.

- Built a reproducible Python 3.13 research environment with uv and PEP 517, implementing Fama-French three-factor OLS, Black-Litterman allocation, and Kelly sizing for portfolio diagnostics.
- Reduced end-of-day analysis to sub-second pivots by normalising Polygon, yfinance, Alpaca, and FRED feeds into DuckDB with Polars and reusable market-regime labels.
- Stress-tested 10,000 Monte Carlo paths with Cornish-Fisher VaR, reporting 5% VaR of -2.56%, Sortino 3.49, Calmar 6.34, and Jensen alpha 0.686; kept the model firewalled from live execution.
- Made the audit operable through a Typer CLI for dashboard, backtest, and environment checks, plus Streamlit and Plotly views that exposed regime, attribution, and risk evidence to the team.
- Hardened the toolchain with pre-commit checks, Ruff, Bandit, Deptry, MkDocs, Doppler-managed secrets, and immutable Appwrite snapshots so results remained reviewable and reproducible.

BWC Adaptive Efficiency Investment Desk

Investment simulation: USD 1 million mandate combining adaptive risk architecture, investment-committee governance, and a decision-focused postmortem.

- Led an eight-person desk through a USD 1 million StockTrak mandate, combining systematic factor selection with discretionary macro overlays informed by Project GANET research across AI, defence, financials, and defensive retail.
- Protected capital during the 2 April tariff shock by activating the Section 5 override, moving into cash and inverse-ETF hedges, and containing the trough at -6.44% versus SPY at -11.72%, a 528-basis-point downside advantage.
- Kept the portfolio within its IPS through 25% single-name limits, drift checks, and regime sleeves calibrated near 1.0 beta for Foundation, 1.1 for Growth, 0.0 for Crisis, and 0.6 for Ceasefire conditions.
- Re-risked after the crisis window using an explicit April playbook rather than discretionary drift, finishing the mandate at -4.37% with 0.63 beta while preserving committee accountability through trade logs and bi-weekly decision diaries.
- The behavioural audit exposed a 28.6% win rate and losses held roughly twice as long as winners; converted the findings into new position-sizing and exit rules.
- Defended the strategy through a 2,600-word investment-committee memo and 20-slide oral defence, earning 422/430 overall, including 147/150 for architecture and logs, 118/120 for strategy and reporting, and 118/120 for the client postmortem.

Bloom - Creative Terminal Visualization

2026 creative-development project: physics-driven terminal bonsai simulation for constrained TTY environments and the Unix ricing community.

- Architected a physics-driven terminal bonsai simulation in Python and Rich, rendering procedural branch growth, wind perturbation, and falling-petal particles at 60 frames per second in constrained TTY environments.
- Implemented Sakura, Wisteria, Autumn, and Winter palettes through HSL interpolation, enabling runtime palette switching with no re-render latency through buffered screen-update cycles.
- Built a deterministic procedural-generation engine for fractal branch topology, parameterising growth angles, bifurcation probability, and bloom density through configurable L-system approximations.
- Engineered real-time petal physics for gravity, air resistance, terminal velocity, and wind-vector deflection, managing more than 200 concurrent entities with $O(n)$ updates in a single-threaded asynchronous loop.
- Exposed interactive wind direction, strength, turbulence, and damping through WASD controls, turning the animation into a responsive terminal experience for the Unix ricing community.
- Maintained sub-50 ms frame latency across Kitty, Alacritty, iTerm2, and Windows Terminal through frame capping, buffered updates, and dirty-region rendering.

- Shipped a reproducible uv and PEP 517 environment with a locked dependency graph, Ruff, Pyright, and pre-commit checks, preserving type safety and preventing dependency drift.

The Bard - Static Publishing Engine

2026 full-stack architecture project within Project GANET, providing an article-first and host-neutral publishing engine for research and knowledge gardens.

- Designed an article-first static publishing engine supporting Obsidian Markdown, hand-authored standalone HTML, and Pandoc-generated single-page HTML under one metadata registry with bidirectional wikilinks and backlink resolution.
- Architected a Jinja2 templating layer around a Readest-derived reader engine, delivering themeable typography, focus-mode reading, and scroll-progress tracking while preserving a portable host-neutral dist artifact.
- Implemented garden-graph visualisation, an atomic Google Drive mirror, and a GitHub Actions deployment pipeline, turning GANET's source material into a reproducible publishing workflow.

Restaurant Operations Platform for Independent Client

Hospiteri was a client project delivering a full-stack operating system for an independent restaurant.

- Delivered a full-stack restaurant platform that unified reservations, menu administration, point of sale, staff cashiering, and inventory control in one operating system.
- Mapped reservations, menu, POS, staff, and inventory requirements into a structured data model and executable workflows before development, reducing ambiguity across the build.
- Converted the initial paid build into a 2-month technical mentorship when the owner asked to learn the codebase, then remained core engineer while transferring day-to-day ownership through documentation and training.

Presentation Format Conversion Platform

Local-first Electron presentation platform supporting format round-tripping, preserved animation intent, and extensible conversion workflows.

- Built bidirectional round-tripping among PPTX, Google Slides, and HTML inside a local-first Electron shell, preserving animation intent through an extensible plug-in layer.
- Reduced format lock-in by separating the conversion core from format adapters, creating a practical path for new import and export targets without rebuilding the desktop application.

Project GANET - Finance Digital Garden and Learning Collective

Hand-curated and opinionated finance knowledge base combining financial models, papers, research and study material with practical guides and roadmaps for people beginning in finance.

- Built the finance equivalent of a digital garden, curating financial models, papers, research material, study notes, opinionated guides, and beginner roadmaps into a public learning collective.
- Created a two-way learning loop with BWC: GANET research informed the investment desk's models and decisions, while BWC trade logs, performance evidence, and postmortems became worked case material inside GANET.
- Engineered an ingest, validate, attribute, and chart flow using reusable Plotly loops for desk-versus-overlay comparisons and three-dimensional implied-volatility surfaces.
- Enriched portfolio evidence with OpenBB, yield-curve data, market calendars, SEC EDGAR filings, news RSS, and macro sources while preserving provenance through Appwrite snapshots.
- Published a frozen static snapshot with zero runtime data dependencies and documented ingestion, provenance, and publishing in MkDocs, keeping GANET stable, auditable, and reusable beyond any one assignment.

Amanuensis: Hindi Speech Recognition

Open-source Hindi speech-recognition project based on Whisper fine-tuning.

- Built and evaluated a Whisper workflow for Hindi and code-switched speech, carrying raw audio through preprocessing and fine-tuning into prerecorded inference.
- Stopped further training on evidence rather than sunk cost after benchmarking model quality and compute demand against the limits of local hardware.

Agentic Workflow and Evaluation Platform

Production-oriented orchestration and evaluation system for tool-using AI agents.

- Engineered an agent orchestration layer with tool routing, structured memory, retry policies, and mandatory human approval for high-risk actions.
- Built an evaluation harness that compared task success, latency, token cost, tool reliability, and recovery across model and prompt changes.

Generative Search Intelligence System

Search and content-intelligence workflow spanning SEO, AEO, and generative discovery.

- Translated search intent into answer-ready GEO and AEO briefs by connecting entity coverage, source quality, structured data, and distribution.
- Designed an experiment dashboard that tied crawl coverage and citations to branded visibility, conversion, and performance change after each content release.

Independent Culinary Development and Kitchen Systems Practice

Structured culinary research and professional kitchen-systems practice spanning repeatable preparation, food safety, fermentation control, and service readiness.

- Improved service readiness through repeatable mise en place routines spanning vegetable preparation, knife work, sauce bases, fermentation, temperature checks, and close-down hygiene.
- Standardised culinary development into a service-ready system built around preparation lists, labelled storage, FIFO rotation, allergen separation, and timed batch plans.

Leased Hotel Turnaround and Operating Model Strategy

Two-month entrepreneurship-through-acquisition turnaround of a loss-making leased hotel property, from operating review and digitisation through revenue redesign, kill-gate decision, and documented offboarding.

- Gave the hotel team a shared turnaround baseline in month one by mapping RevPAR, occupancy, staffing, guest satisfaction, asset condition, and guest or staff friction.
- Reduced day-to-day operating friction in month two by designing mobile-first front-desk and staff workflows with the supporting technical architecture and infrastructure.
- Raised month-on-month occupancy above the inherited baseline by turning RevPAR, staffing, guest-satisfaction, and occupancy analysis into a mixed-revenue operating model.
- Applied the two-month profitability kill gate, returned the property to prior management, and preserved the turnaround work in an offboarding trail and future-lease pitchbook.

Culinary Operations Rotation

Commercial back-of-house and front-of-house rotations across multiple London restaurants and live service environments.

- Executed knife work, sauce bases, and timed batches against live service windows while maintaining FIFO, allergen, and temperature controls.
- Maintained saucier control from production through handover, protecting reduction timing, multi-sauce consistency, holding temperature, and close-down hygiene.
- Protected order accuracy under peak volume by coordinating service, resolving guest issues immediately, and shadowing the maitre d' through front-of-house handover.

Danone YoPRO APEX Diagnostic

Team strategy case: board-level corporate entrepreneurship playbook for a modelled Danone precision-nutrition platform.

- Co-authored a 2,989-word board playbook recommending YoPRO APEX as an integrated precision-nutrition platform built around a bioactive formulation, dual-compartment pack, and AI data layer.
- Turned the diagnostic into a client-facing HTML presentation that made the strategic choice, portfolio architecture, and governance gates legible to an executive audience.
- Structured Core YoPRO, APEX, and Huel as a governed portfolio with explicit go, pivot, and kill criteria at each Stage-Gate decision.
- Modelled a phased UK pilot at GBP 3.20 per 200 g pot to test a roughly 30% premium against the 1.13 million-entry London Marathon market.

IHCL/Taj Hotels Corporate Innovation Diagnostic

Independent strategy case: Project Bifocal examined how Taj could scale while protecting its service heritage.

- Mapped Taj's service-capacity constraint across 129 properties and the 150-property scaling threshold, then built Project Bifocal around that gap.
- Allocated a proposed INR 1,500 crore venture envelope through the Three Box framework and fuzzy gates tied to repeat purchase, NPS, contribution margin, and kill thresholds.
- Designed an INR 150 crore, five-year Tajness Transmission Academy to prepare 2,000 to 2,500 staff annually and protect service consistency during expansion.
- Generated the 20-slide executive deck programmatically from the end-to-end diagnostic, preserving analytical consistency across the board narrative.

NIRMANA Design Capstone

Cross-format design-learning system built around a practitioner toolkit, curriculum, media, and spatial experience.

- Created NIRMANA through 4 brief iterations, building a design-learning system around 75 gamified questions that help practitioners recover observation, practice, repetition, and discourse.
- Extended the critique of design education into 7 connected formats: curriculum, essays, short film, music score, microsite, documentation, and spatial manifestation.
- Built 2 companion interfaces - a public microsite and a Snaptrude spatial prototype - connecting the digital card system to its proposed physical exhibition environment.

US Defence Drone Spinout Commercialisation Assessment

Venture-readiness and commercialisation assessment for a US-based academic team's defence-related drone technology spinout.

- Assessed a US academic drone spinout using KTH Innovation-style venture-readiness criteria, testing technical feasibility, commercial potential, company formation, and execution risk.
- Flagged material governance and business-law risks across university IP ownership, licensing, founder roles, vesting, cap-table formation, funding-round dilution, and defence regulation.
- Mapped a gated commercialisation path from technical validation and founder agreements through defence-market entry, financing readiness, and the next spinout decision.

Duve Venture Diligence and Industry Tearsheet

Investor-side diligence exercise for an anonymised London VC evaluating Tel Aviv-based Duve at the bridge between seed and Series A.

- Delivered an industry tearsheet and diligence pack for an anonymised London VC evaluating Tel Aviv-based Duve during its bridge between seed and Series A.
- Tested Duve's investment case against PMS dependencies, hotel-operator workflows, rollout friction, customer concentration, downside exposure, and strategic synergies.
- Framed Duve as a conditional follow-up opportunity rather than a clean proceed/pass decision, identifying PMS integration dependency, rollout friction, and customer concentration as the key diligence gates before conviction.

AstroPatshala AARRR Growth Strategy

Pirate-funnel growth strategy for AstroPatshala's consulting, education, and research offers.

- Mapped the AARRR funnel across AstroPatshala's consulting, education, and research offers, linking acquisition, activation, retention, referral, and revenue to concrete operating choices.
- Turned the funnel into an operating roadmap spanning channel tests, onboarding, repeat-service triggers, referral loops, and pricing actions.

JEKI Cross-Channel Campaign for AfriKids

Client-connected project for AfriKids: alternative-reality campaign designed to convert play, discovery, and community into awareness and donations.

- Designed JEKI as a biannual alternative-reality campaign spanning 5 participation channels - web puzzles, city clues, pop-ups, Discord, and merchandise - to convert play into charity awareness and donations.
- Built 4 launch assets - the JEKI and Yago website prototype, a dual-brand system, poster series, and cross-channel journey - linking public discovery to an invite-only creative community.

Redshaw Advisors Brand Strategy and Campaign Kit

Independent client engagement redesigning Redshaw Advisors' market positioning, brand identity, and campaign model from carbon-market diligence and ESG research.

- Redesigned Redshaw Advisors' marketing and positioning model by turning carbon-credit diligence and ESG research into a clearer brand identity, audience proposition, and campaign architecture.
- Paired the repositioning with guerrilla campaign concepts and a reusable creative kit designed to make specialist carbon-market expertise more legible and convert attention into qualified interest.
- Earned client commendation for the clarity and cohesion of the final strategic platform, brand direction, and campaign system.

Carbon Forward Conference Lead-Magnet Design

Client design engagement for the Carbon Forward conference.

- Served as deputy team lead for a real-client Carbon Forward engagement, steering research and delivery toward a conference lead magnet grounded in carbon-market diligence.
- Accelerated delivery by reusing validated carbon-credit and ESG research from the Redshaw engagement while rebuilding the message for Carbon Forward's audience and lead-generation objective.

London Airbnb Pricing and Regression Models

Team analytics case using a London Airbnb dataset, in which I owned pricing-model development.

- Led the project team as primary developer and designer, owning the analytical workflow from data preparation and model design through interpretation and presentation.
- Developed a log-linear pricing model on 6,319 London Airbnb listings and 13 variables, explaining 50% of price variance and placing 73% of predictions within 1 price band.
- Quantified key price effects, including 44% lower prices for private rooms, 58% lower for shared rooms, 12% higher per additional guest, and 8% higher per bedroom.
- Retained 457 legitimate market outliers, representing 7.2% of listings, and used log transformation instead of deletion to preserve market heterogeneity while stabilising variance.

Project OULM - One YMCA Youth Community Platform

Client-connected digital-to-physical youth service ecosystem designed for One YMCA across Bedfordshire, Hertfordshire, and Luton.

- Led a 5-person team in designing OULM across 3 service areas - Bedfordshire, Hertfordshire, and Luton - moving young people from digital discovery and booking into Welcome Areas and the Woodland Cafe.

- Shifted the growth model away from daily active users and toward offline belonging, with intentional friction and physical verification for under-16 access.
- Delivered 6 connected outputs - technical MVP, identity, design tokens, skeleton states, posters, and venue touchpoints - as a safeguarding-aware service ecosystem.
- Connected online pathways to 2 physical destinations - Welcome Areas and the Woodland Cafe - so discovery, safeguarding, and community participation resolve in real places.

Hospitality Craft and Innovation Article

Analytical article examining the decline of hospitality craft as an innovation challenge.

- Published an evidence-led argument connecting service quality, workforce capability, and operating-model design as mutually reinforcing levers for hospitality innovation.
- Argued that hospitality should invest in training, service judgement, and repeatable execution rather than treating innovation as technology adoption alone.

Family Governance and Office Access Model

Karta is active strategic research and proposal work on family governance, HUF coordination, and family-office access beyond ultra-high-net-worth households.

- Designed a coordinating family-governance and decision-rights framework that clarified ownership, authority, information flow, accountability, and the boundary between family, business, and investment decisions.
- Modelled how traditional HUF families could work with a single-family office or shared multi-family office, preserving cultural and legal realities while professionalising investment, reporting, succession, and adviser coordination.
- Unbundled family-office services into governance, reporting, diligence, succession, and specialist-adviser modules for families in Tier 2 and Tier 3 cities.
- Turned the research into active proposals and operating cadences for investment approvals, related-party decisions, succession, conflict escalation, record keeping, and adviser handoffs.

PACA Applied Food Science

Controlled small-batch food-science practice using Monte Carlo modelling to test ingredient combinations and fermentation variables.

- Used Monte Carlo analysis to compare ingredient combinations and fermentation conditions within a controlled small-batch culinary dataset.
- Used model output to build a controlled test matrix, narrowing the combinations taken forward for physical preparation and fermentation observation.

PACAKATVA Culinary Data Platform

Bloomberg-Terminal-style culinary intelligence and community platform combining proprietary data, provenance, expert curation, and a third-space concept for chefs.

- Extended PACA into a Bloomberg-Terminal-style culinary intelligence platform combining cookbook data, live wholesale provenance, and expert curation.
- Positioned a London pilot for food-curious professionals aged 24 to 38 to validate decision utility before committing to full real-time data infrastructure.
- Built the product narrative, data architecture, curation model, and terminal-style prototype around a human-signal and proprietary-data moat plus a community third-space proposition for chefs.

Personal Knowledge Management System

Sage is a professional-grade system for diary keeping and the capture, storage, retrieval, and connection of research, experience, decisions, and learning.

- Maintained the working memory for 53 projects across four foundational domains, connecting daily diaries, research, source evidence, decisions, and final deliverables in one professional knowledge system.
- Designed a repeatable capture, store, connect, retrieve, and reuse framework so new work could build on prior learning without losing the context behind a decision.
- Executed a major archive sprint within Sage, reorganising approximately 47,000 files across 16 GB with an MPLP-inspired method and a searchable SQLite manifest.
- Made the archive auditable without depending on folder memory by preserving file identity, location, provenance, and retrieval metadata in the manifest.

Design Digital Garden

Design-thinking knowledge base for designers who approach the subject from business-first principles, combining an opinionated account of my design practice with a public archive of studies and working knowledge.

- Built a design-thinking knowledge base that translates business-first principles into practical design decisions, connecting commercial context, user needs, systems constraints, and creative execution.
- Curated an opinionated record of my design work, experience, and learning, exposing the reasoning, process notes, critiques, and connections behind projects rather than showing only polished outcomes.
- Published studies, frameworks, references, and project documentation through a non-linear information architecture that lets designers move between foundational concepts and applied case work.

EcoLithify Sustainable Tableware Concept

Sustainable tableware concept using marble-slurry waste from Jaipur's stone industry.

- Applied circular-design principles to Jaipur marble slurry, connecting material feasibility and manufacturing economics to a hospitality-facing tableware position.
- Set material prototyping and supplier validation as preconditions for scale-up, controlling manufacturing and food-contact risk.

Crestara Smart-Room Digital Twin and Privacy-First IoT Platform

Venture design case: hotel-room digital twin expanded into a privacy-first modular IoT concept with PMS integration and tiered deployment economics.

- Built a hotel-room digital twin from the ground up, linking occupancy, energy use, guest controls, maintenance demand, and room-level cost into a decision model.
- Expanded the prototype into Crestara, integrating PMS, networking, television, and room controls while processing guest data on site.
- Engineered privacy into the architecture through checkout-triggered deletion, guest-requested erasure, transparent audits, local processing, and removable devices.
- Extended the digital twin into an underwriting model testing capital cost, occupancy, energy, maintenance, subscription revenue, and payback across deployment scenarios.
- Modelled a tiered launch with GBP 65,000 initial capital, hardware priced from GBP 329 to GBP 999, recurring subscriptions, and GBP 800 annual maintenance contracts.

Global Development and Climate Hypothesis Testing

Sainen was an independent research case testing economic development against emissions and net-zero policy commitments.

- Merged GDP, emissions, and net-zero datasets into 10,199 country-year observations, then built a stratified 1,800-record sample spanning 191 countries.
- Tested 2 policy hypotheses with correlation, ANOVA, and chi-square methods, finding significant GDP associations with emissions and legal climate commitments at $p < 0.001$.
- Translated the results into a supplier-risk framework for carbon-border exposure, separating 3 market categories: legally committed, policy-only, and uncommitted.

Noritake Hospitality Tableware Positioning Strategy

Independent strategy case repositioning Noritake's tabletop capability for Asian hospitality partnerships.

- Mapped Noritake's JPY 147 million tabletop loss against Asian hospitality whitespace using Blue Ocean Strategy and strategic-group analysis.
- Modelled 15% to 25% premium pricing across 40% to 50% of tabletop revenue, projecting USD 54 million to USD 90 million in incremental revenue from repositioning.
- Built the repositioning around a hotel configurator, 24 to 36 month partnership contracts, co-marketing, and sustainability-led supply-chain optimisation.

Toast Hotel F&B Expansion Strategy

Independent strategy case using a resource-based blueprint to extend Toast into hotel food and beverage operations.

- Used VRIO to identify Toast's deployment capability across 156,000 locations and USD 186 billion in annual payment volume as the defensible base for a hotel F&B vertical.
- Modelled a hotel F&B vertical for 50 to 200-property chains, projecting USD 120,000 ACV, an 11.9:1 LTV/CAC ratio, and USD 6.82 billion incremental 10-year NPV.
- Mapped a 100-day enterprise entry plan across pilot agreements, a hotel sales team, PMS partnerships, and explicit conversion and churn gates.

Octopus Energy Battery Integration Strategy

Team strategy case for integrating utility-scale storage with Octopus Energy's technology platform.

- Recommended Option D, a GBP 150 million backward-integration move into utility-scale batteries connected to Kraken and PowerGuard, after testing industry pressure with Porter's Five Forces and Kraken's roughly 40% cost-to-serve advantage.
- Prioritised the 200 to 400 MW battery path over three alternatives using RICE and scenario planning; it scored 253 and modelled GBP 160 million to GBP 220 million of impact.
- Turned the recommendation into a 24-month roadmap with gates above 20% IRR, GBP 40 million in annual grid contracts, 93% availability, and pre-agreed pivot or partner contingencies.

Walmart Capital Budgeting and Future-Proofing Model

Team capital-allocation case evaluating a USD 100.76 billion Walmart automation, store technology, and EV infrastructure programme.

- Co-built a capital-budgeting model for a USD 100.76 billion programme spanning automated fulfilment, store technology, and EV infrastructure.

- Calculated a 7.8% WACC, USD 164.05 billion NPV, 15.7% IRR, 2.83 profitability index, and discounted payback under 9 months using base-case operating and financing assumptions.
- Stress-tested the investment across plus or minus 20% revenue, WACC, and cost scenarios and translated the outputs into capital-allocation recommendations.

Walmart Pfand Circular Retail Strategy

Team future-proofing case for Walmart reverse-vending, rewards, and accessible store operations.

- Designed a Walmart+ reverse-vending roadmap with accessible kiosk UX, instant rewards, fraud detection, and a modelled scale phase covering more than 2,000 stores.
- Set pilot targets of 50% fewer interface errors, 40% less manual intervention, 15% higher satisfaction, and 25% higher effective kiosk sales.

Samsung and Huawei Strategic Resilience Analysis

Independent strategy case comparing Samsung and Huawei on technology, ESG, geopolitical exposure, and supply-chain resilience.

- Compared Samsung and Huawei using PitchBook and company data, connecting semiconductor depth and ESG commitments to geopolitical exposure and supply-chain resilience.
- Built a risk register and opportunity map linking 3 nm manufacturing, USD 18.8 billion R&D, sanctions, circularity, and domestic sourcing to future market access.

Personal Brand and Digital Presence

Raconteur is a personal narrative, brand, and publishing system spanning major channels and owned web properties.

- Unified positioning across LinkedIn, Instagram, and kartavya.tech through a shared visual identity, headline, and content architecture.
- Turned the owned website into the source of truth for projects and long-form evidence, giving shorter social profiles a consistent path back to substantive work.

Education

Hult International Business School - London, UK

Expected May 2027

Undergraduate Business School | Bachelor of Business Administration

Double Major: Entrepreneurship and Design

3.76/4.00 | First Honors (UK First-class standing)

Relevant Coursework: Acting 101 Voice, Body, and Character; Business Strategy; Start-up Challenge; Social Impact Challenge; Future Proofing Challenge; Corporate Entrepreneurship; Creativity; Design 1; Design Thinking in Practice; Fundamentals of Business Analytics; Building Next-Generation Startups; Design Thinking in Web Design

Honors: Dean's List for consecutive terms; Global campus winner, SDG-focused Social Impact Challenge

Skills

Backend: Python, Flask, Django, FastAPI, JavaScript, HTML/CSS, SQL, Git/GitHub

Data: MySQL, PostgreSQL, SQLite, Redis, Excel, OpenPyXL, Power BI

AI / ML: TensorFlow, PyTorch, Whisper, speech-model fine-tuning, real-time AI voice systems

Systems: Linux Systems Administration, Docker, network routing, firewall traversal, SSH deployment

Product / Design: Figma, Snaptrude, product strategy, systems design, service design

Founder / Venture: Founder storytelling, GTM playbooks, MVP development, Python, Native web development

Culinary Consulting: Menu engineering, Restaurant analytics, Food-tech product design, OCR data digitisation, Excel and OpenPyXL, Service design

Kitchen Craft: Mise en place, Knife skills, Sauce preparation, Food safety and allergens, FIFO stock rotation, Service timing

BOH Operations: Production planning, Inventory control, FIFO rotation, Shift handovers, Kitchen reconciliation, Excel and OpenPyXL

FOH / Reception: Guest services, Reservations, Reception handovers, Cash handling, Service recovery, Order accuracy

Languages: English (Fluent), Hindi (Fluent)

Certifications: PitchBook Certification Suite; Selected Coursework in Business, Technology, and Analytics

Work authorization: Authorized to work up to 20 hours per week in the United Kingdom under current visa conditions and full-time in India. Open to relocation and employer sponsorship for full-time roles elsewhere.